House Price Prediction – Project Overview

Introduction

The aim of this project was to design a machine learning model that could estimate house prices using property attributes such as plot size, number of bedrooms and bathrooms, number of floors, parking facilities, furnishing status, and other amenities. Reliable price prediction models are valuable for buyers, sellers, and real estate firms to make smarter, data-driven decisions.

Data Exploration and Preparation

The dataset contained 545 entries with 13 features. Using Python and Pandas, the data was inspected to understand its structure, identify the target variable, and check for inconsistencies. The dataset was clean—no missing values or duplicates were found. To prepare categorical features for modeling, one-hot encoding was applied, ensuring compatibility with regression algorithms.

Model Building

The dataset was split into training (80%) and testing (20%) subsets. Two regression approaches were implemented and compared:

1. **Linear Regression**
2. **Random Forest Regressor**

Performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Results

Linear Regression

- MAE: 970,043
- RMSE: 1,324,507
- R^2: 0.653

Random Forest Regressor

- MAE: 1,021,546

- RMSE: 1,400,566
- R^2: 0.612

Linear Regression outperformed Random Forest in this case, achieving a higher R^2 and lower error values.

Insights

Feature importance analysis highlighted the following as the strongest predictors of house price:

- **Area** (46.84%)
- **Bathrooms** (15.15%)
- **Air Conditioning** (6.27%)
- **Parking** (5.75%)
- **Stories** (5.71%)

This suggests that larger homes with more amenities tend to command higher prices.

Conclusion

The project demonstrated how machine learning can be applied to real estate pricing. Linear Regression proved to be the most effective model for this dataset, explaining about 65% of the price variation. The findings emphasize the role of property size and key amenities in determining value, offering practical insights for real estate professionals and buyers alike.